SENSOR DRIVEN ADAPTIVE SANITATION PROTOCOL

Component Overview, Bill of Materials & Project Cost Analysis

IIMSTC | Cohort 2 — IoT & Embedded Systems | NodeMCU ESP32-S Platform

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Project Overview

Smart IoT Sanitation Monitoring System

10
Components

7
Project Phases

₹3,603
Total Cost

What is this project?

The Sensor Driven Adaptive Sanitation Protocol is an IoT-based hygiene monitoring system built on the **NodeMCU ESP32-S** platform. It uses gas, IR, and moisture sensors to detect and alert on sanitation issues in real time — automating exhaust control, providing staff alerts via buzzer and LCD, and enabling keypad-authenticated acknowledgment.

- NodeMCU ESP32-S
- MQ-136
- MQ-137
- IR Sensor
- LCD Display
- Relay
- Buzzer

Components Overview

10 components across 7 development phases

NodeMCU ESP32-S

Main microcontroller — all phases

MQ-136 Gas Sensor

H_2S detection — Phase 4

MQ-137 Gas Sensor

NH_3 detection — Phase 4

IR Sensor (FC-51)

Occupancy detection — Phase 3

Moisture Sensor (YL-69)

Floor wetness — Phase 4

DHT22 Sensor

Temp & humidity monitoring

16×2 LCD (I^2C)

System status display — Phases 2–6

4×3 Matrix Keypad

Staff PIN verification — Phase 6

Relay Module

Fan/appliance control — Phase 5

Buzzer

Audible alerts — Phase 5

Core Platform & New Sensor

NodeMCU ESP32-S + DHT22 Humidity & Temperature Sensor

NodeMCU ESP32-S — The Brain

Part No.	NodeMCU ESP32-S (38-pin)
CPU	Xtensa LX6 240MHz
GPIO	30+ pins
ADC	12-bit, multiple ch.
Flash	4MB
Wi-Fi	802.11 b/g/n
Voltage	3.3V / 5V via VIN
Phase	All Phases

DHT22 (AM2302) — NEW SENSOR

Part No.	DHT22 / AM2302
Temp Range	-40°C to +80°C
Temp Accuracy	±0.5°C
Humidity Range	0–100% RH
Humidity Accuracy	±2–5% RH
Output	Single-wire digital
Voltage	3.3V – 5V
Use Case	Env. comfort monitoring

Gas & Environmental Sensors

Phase 4 — MQ-136 · MQ-137 · DHT22 · Moisture Sensor

MQ-136 — Hydrogen Sulfide (H_2S)

- Range: 1–200 ppm
- Detects H_2S — the 'rotten egg' odor in toilets
- Analog output read by ESP32 ADC
- Triggers exhaust fan & alert above threshold
- Heats up ~20s on power-on for calibration

MQ-137 — Ammonia (NH_3)

- Range: 5–500 ppm
- Detects NH_3 — urine / cleaning chemical odor
- Dual-sensor setup for comprehensive air quality
- Works alongside MQ-136 for multi-gas monitoring
- High NH_3 level can indicate cleaning is needed

DHT22 — Temp & Humidity

- Temperature: -40°C to +80°C (±0.5°C)
- Humidity: 0–100% RH (±2–5%)
- Single-wire digital protocol
- Monitors bathroom comfort conditions
- Triggers alerts on abnormal temp/humidity

Moisture Sensor (YL-69)

- Detects water or liquid spills on toilet floor
- Analog output read by ESP32 ADC
- Triggers cleaning alert if wetness exceeds threshold
- Helps prevent slip hazards
- Works alongside gas sensors for full hygiene check

Interface, Alerts & Control

LCD Display · Keypad · Relay · Buzzer · IR Sensor

16×2 LCD Display (JHD162A)

Shows system status across all 7 phases. Displays sensor readings (gas ppm, moisture %). Prompts keypad PIN entry for staff verification. I^2C backpack reduces wiring to just 2 pins (SDA/SCL).

4×3 Matrix Keypad

12 keys (0–9, *, #) for PIN entry in Phase 6. Staff enters verification PIN to acknowledge alerts. Prevents false alarm dismissal by unauthorized users. Membrane design — waterproof and easy to clean.

Relay Module (SRD-05VDC)

Switches exhaust fan ON when gas threshold exceeded. ESP32 3.3V GPIO safely drives relay via optocoupler. Controls 230V AC loads with low-voltage signal.

Active Piezo Buzzer (5V)

Emits audible alert when gas/moisture alarm triggers. Active buzzer — only needs digital HIGH to sound. Can be silenced after keypad PIN verification in Phase 6.

IR Sensor (FC-51 / TCRT5000)

Detects entry and exit events at toilet door. Increments/decrements occupancy counter. Prevents false alerts when stall is unoccupied. Adjustable sensitivity via onboard potentiometer.

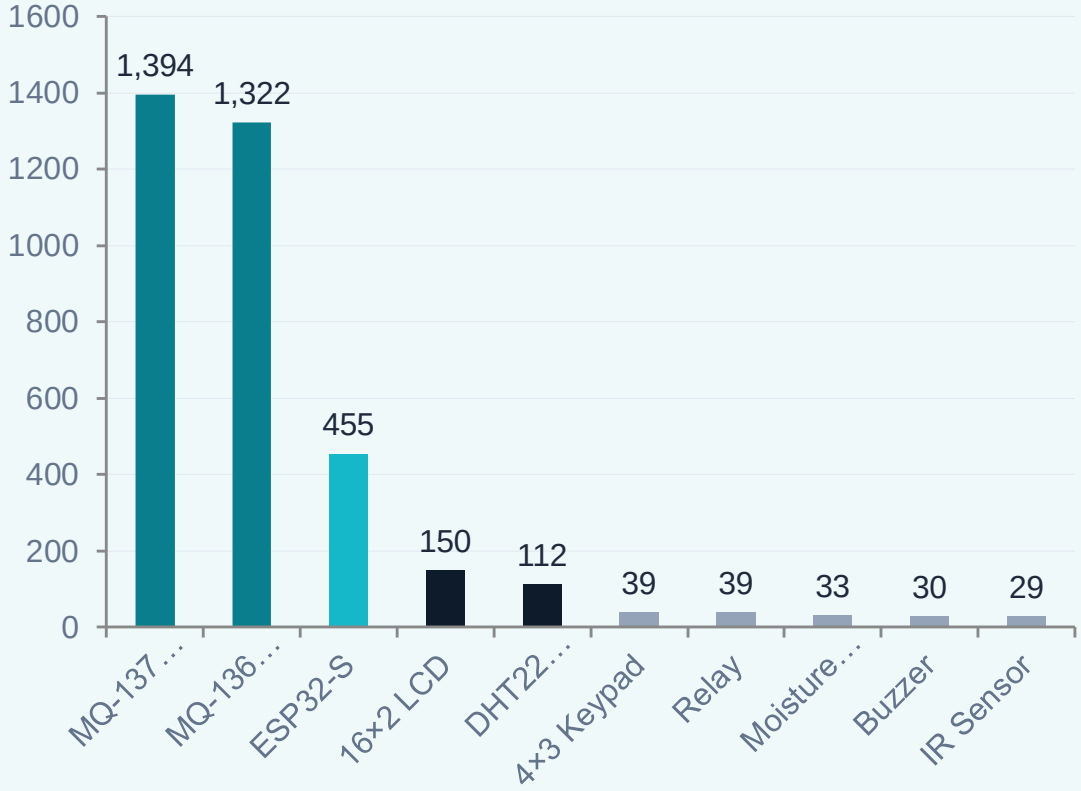
Bill of Materials

Complete component list with pricing — April 2026

S. No	Component	Part No. / Model	Phase	Qty	Unit Cost	Total Cost
1	NodeMCU ESP32-S	NodeMCU ESP32-S (38-pin)	All	1	₹455	**₹455**
2	IR Sensor	FC-51 / TCRT5000	Phase 3	1	₹29	**₹29**
3	Gas Sensor H_2S	MQ-136	Phase 4	1	₹1,322	**₹1,322**
4	Gas Sensor NH_3	MQ-137	Phase 4	1	₹1,394	**₹1,394**
5	Moisture Sensor	YL-69 / HL-69	Phase 4	1	₹33	**₹33**
6	16×2 LCD Display	JHD162A + I^2C Backpack	Phases 2–6	1	₹150	**₹150**
7	4×3 Matrix Keypad	Generic Membrane 4×3	Phase 6	1	₹39	**₹39**
8	Relay Module	SRD-05VDC-SL-C (1ch)	Phase 5	1	₹39	**₹39**
9	Buzzer	Active Piezo Buzzer 5V	Phase 5	1	₹30	**₹30**
10	DHT22 Sensor	DHT22 / AM2302	Phase 4	1	₹112	**₹112**
	GRAND TOTAL					**₹3,603**

Cost Breakdown Analysis

Visual breakdown of component costs — Total: ₹3,603



Cost Distribution

Gas Sensors (MQ-136+137)
₹2,716 75.4%

DHT22 Sensor
₹112 3.1%

Microcontroller (ESP32-S)
₹455 12.6%

Display + Interface
₹228 6.3%

Control & Alerts
₹92 2.6%

7-Phase Development Roadmap

Progressive build from core setup to full alerting system

Ph 1	Ph 2	Ph 3	Ph 4	Ph 5	Ph 6	Ph 7
1	2	3	4	5	6	7
Environment Setup	**LCD Integration**	**IR Occupancy**	**Sensor Suite**	**Alert & Control**	**PIN Keypad Auth**	**Cloud/Dashboard**
ESP32-S setup, IDE config, GPIO test	I^2C LCD wiring, display text/status	Door sensor, counter logic	MQ-136, MQ-137, DHT22, Moisture	Relay fan control, Buzzer alerts	Staff PIN verification system	Wi-Fi logging, optional RTC
			★ KEY PHASE			

Summary & Key Takeaways

- 10 components selected across 7 development phases
- DHT22 added for temperature & humidity environmental monitoring
- Dual gas sensing (MQ-136 + MQ-137) for comprehensive air quality coverage
- Total project cost estimated at ₹3,603 (approx. market rates, April 2026)
- ESP32-S provides built-in Wi-Fi for optional cloud logging & dashboards
- Breadboard, PCB, jumper wires not included (already available with team)